📘 Assignment: Designing an Efficient Fine-Tuned LLM

Objective

You are tasked with designing a **medical domain assistant** using a 7B open-source LLM under a **16GB T4 GPU constraint**.

Part 1 — Memory & Feasibility

Q1. Can We Do Full Fine-Tuning?

1. Calculate memory required for:
    - 7B model in FP16
    - Include weights only
2. Briefly explain what extra memory is required during training:
    - Gradients
    - Optimizer states
3. Conclude: Why is full fine-tuning impractical on 16GB GPU?

Part 2 — PEFT Strategy Design

Choose ONE:

- LoRA
- QLoRA
- Adapters
- Prefix Tuning

Explain:

- Why you chose it
- Approximate % of trainable parameters
- Memory benefits
- One limitation

Part 3 — 4-bit vs 8-bit Decision

Assume you are using quantization.

Compare:

- Memory usage difference (7B model)
- Training stability
- Accuracy trade-offs
- When to use 4-bit vs 8-bit

Conclude which you would choose for:

- Training
- Production inference

Part 4 — Multi-Adapter Architecture

You must serve:

- Medical assistant
- Fitness assistant

Using:

- One base 7B model
- Two LoRA adapters

Explain:

- How adapter swapping works
- Why this is memory efficient
- How this helps scale to more domains

Part 5 — Failure Mode Analysis

Briefly explain:

1. Overfitting
2. Catastrophic forgetting
3. Data quality bias

For each:

- Why it happens
- One way to detect it
- One way to prevent it

(Keep concise.)

Deliverables

Submit one PDF (6 pages max) including:

- Memory calculation
- PEFT design decision
- Quantization comparison
- Architecture explanation
- Failure mode analysis